Abstract Classes and Interfaces

CSE 114: Introduction to Object-Oriented Programming

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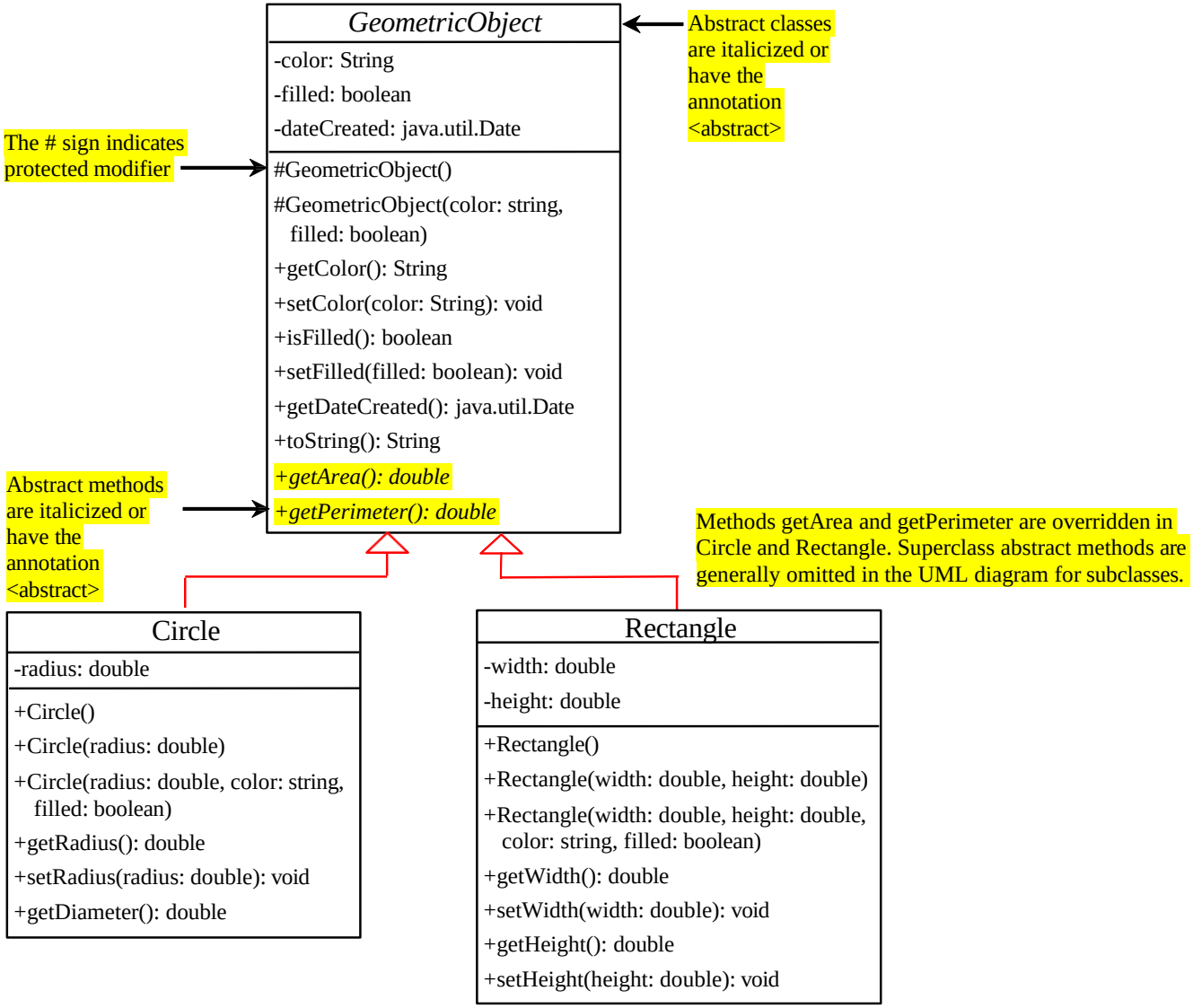
Abstract Classes and Abstract Methods



```java
public abstract class GeometricObject {
  private String color = "white";
  private boolean filled;
  private java.util.Date dateCreated;
  protected GeometricObject() {
    dateCreated = new java.util.Date();
  }
  protected GeometricObject(String color, boolean filled) {
    dateCreated = new java.util.Date();
    this.color = color;
    this.filled = filled;
  }
  public String getColor() {   return color;  }
  public void setColor(String color) {  this.color = color;  }
  public boolean isFilled() {   return filled;  }
  public void setFilled(boolean filled) {  this.filled = filled;  }
  public java.util.Date getDateCreated() {   return dateCreated;  }
  public String toString() {
    return "created on " + dateCreated + "\ncolor: " + color +
      " and filled: " + filled;
  }
  /** Abstract method getArea */
  public abstract double getArea();
  /** Abstract method getPerimeter */
  public abstract double getPerimeter();
}
```

```java
public class Circle extends GeometricObject {
  private double radius;
  public Circle() {    }
  public Circle(double radius) {
    this.radius = radius;
  }
  public double getRadius() {
    return radius;
  }
  public void setRadius(double radius) {
    this.radius = radius;
  }
  public double getArea() {
    return radius * radius * Math.PI;
  }
  public double getPerimeter() {
    return 2 * radius * Math.PI;
  }
  public double getDiameter() {
    return 2 * radius;
  }
}
```

```java
public class Rectangle extends GeometricObject {
  private double width;
  private double height;
  public Rectangle() {
    // super();
  }
  public Rectangle(double width, double height) {
    this();
    this.width = width;
    this.height = height;
  }
  public Rectangle(double width, double height, String color,
        boolean filled) {
    super(color,filled);
    this.width = width;
    this.height = height;
  }
  public double getWidth() {     return width;   }
  public void setWidth(double width) {      this.width = width;   }
  public double getHeight() {     return height;   }
  public void setHeight(double height) {     this.height = height;   }
  public double getArea() {
    return width * height;
  }
  public double getPerimeter() {
    return 2 * (width + height);
  }
}
```

```java
public class TestGeometricObject1 {
  public static void main(String[] args) {
    // Declare and initialize two geometric objects
    GeometricObject geoObject1 = new Circle(5);
    GeometricObject geoObject2 = new Rectangle(5, 3);
    // Display circle
    displayGeometricObject(geoObject1);
    // Display rectangle
    displayGeometricObject(geoObject2);
    System.out.println("The two objects have the same area? " +
       equalArea(geoObject1, geoObject2));
  }

  /** A method for displaying a geometric object */
  public static void displayGeometricObject(GeometricObject object) {
    System.out.println(object); // object.toString()
    System.out.println("The area is " + object.getArea());
    System.out.println("The perimeter is " + object.getPerimeter());
  }

  /** A method for comparing the areas of two geometric objects */
  public static boolean equalArea(GeometricObject object1,
      GeometricObject object2) {
    return object1.getArea() == object2.getArea();
  }
}
```

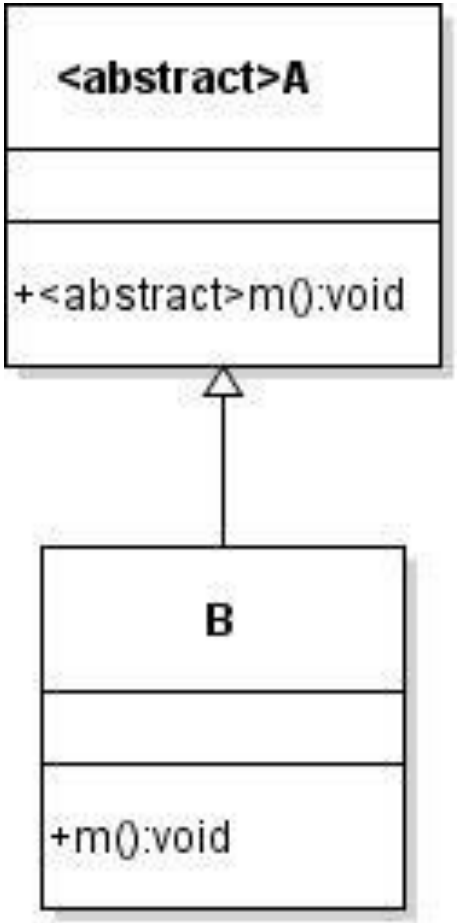
abstract methods in *abstract* classes

- An abstract method can only be contained in an abstract class.

subclasses of *abstract* classes

- In a nonabstract (a.k.a., concrete) subclass extended from an abstract super-class, all the abstract methods MUST be implemented.

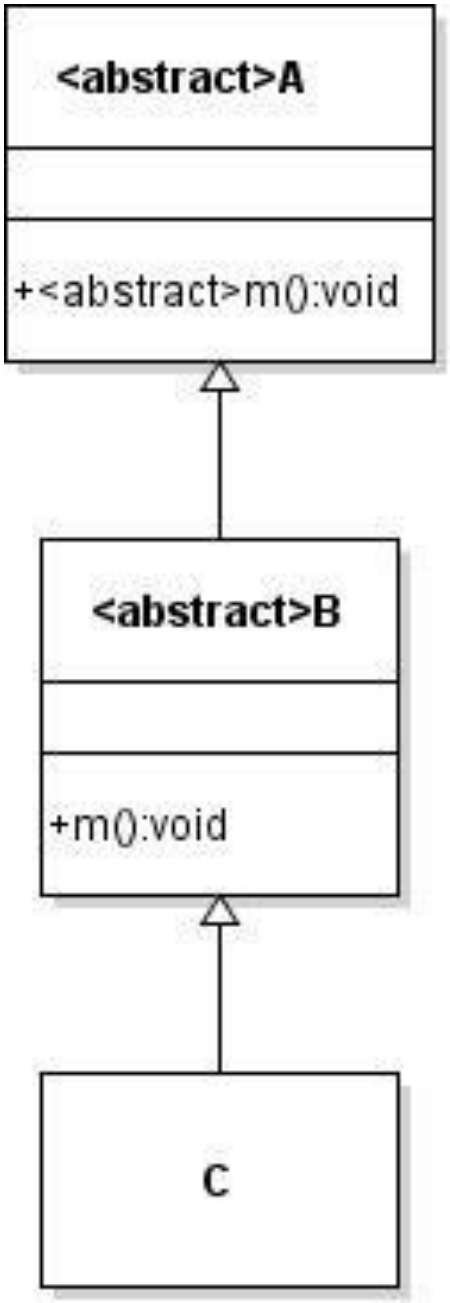
```
abstract class A {
  abstract void m();
}
class B extends A {
  void m(){
  }
}
```



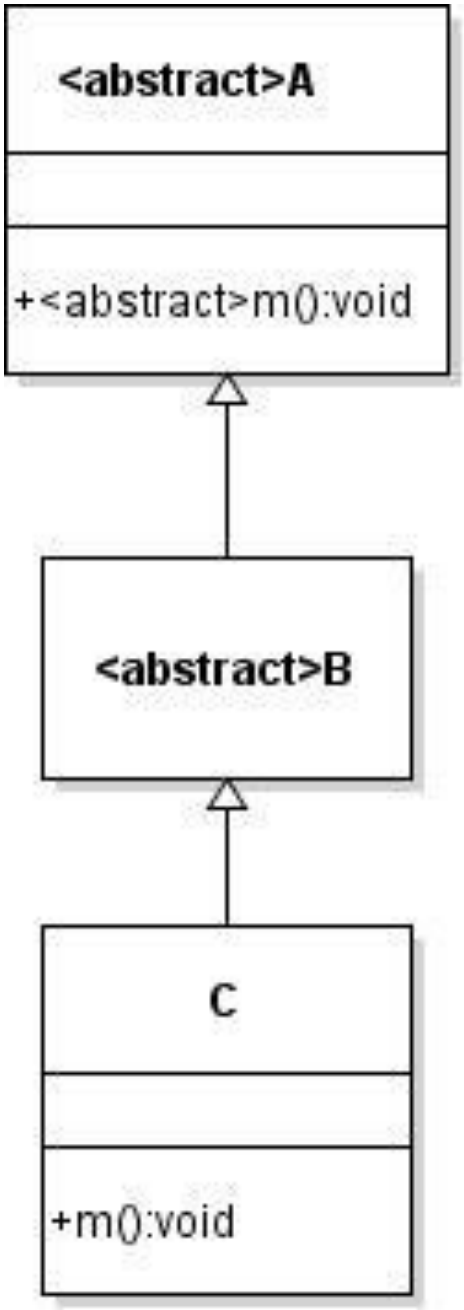
subclasses <u>of</u> *abstract* classes

- In an abstract subclass extended from an abstract super-class, we can choose:
  - to implement the inherited abstract methods OR
  - to postpone the constraint to implement the abstract methods to its nonabstract subclasses.

```
abstract class A {
    abstract void m();
}

abstract class B extends A {
    void m(){
    }
}

class C extends B {

}
```

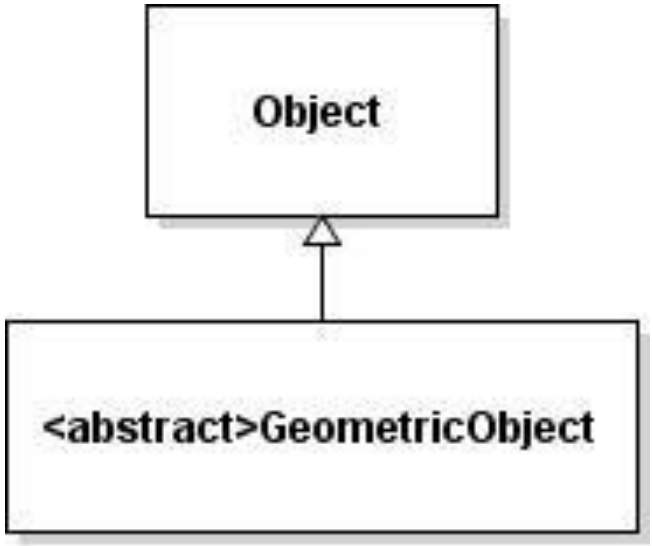


```
abstract class A{
    abstract void m();
}

abstract class B extends A{

}

public class C extends B {
    void m(){
    }
}
```



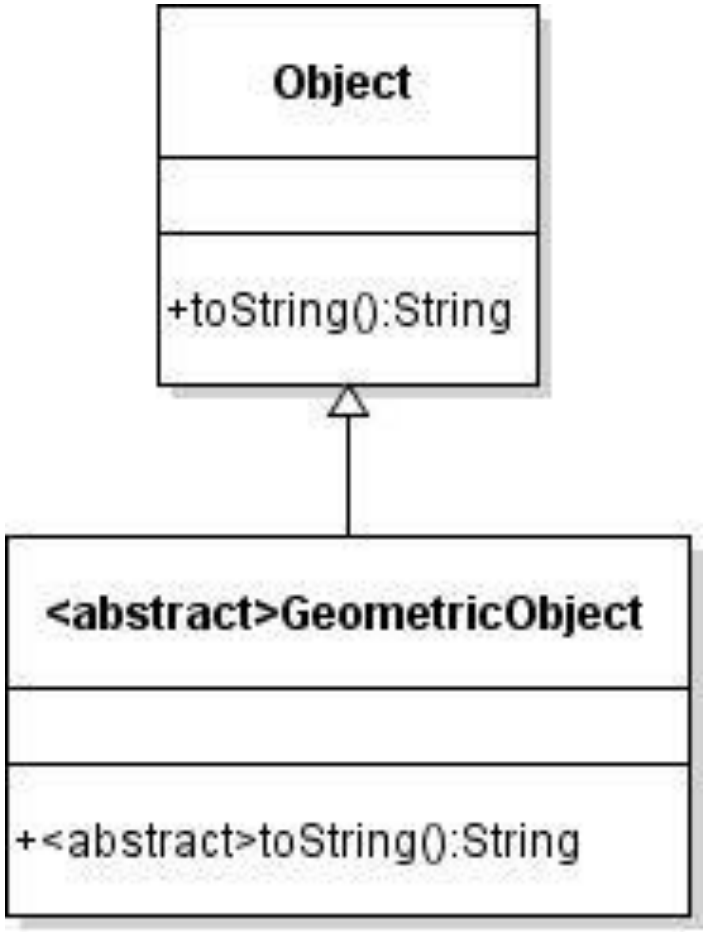
abstract classes

- A subclass can be abstract even if its superclass is concrete.
  - For example, the **`Object`** class is concrete, but a subclass, **`GeometricObject`**, is abstract



abstract classes

- A subclass can override a method from its concrete superclass to define it ***abstract***
  - useful when we want to **force** its subclasses to implement that method, or
  - the implementation of the method in the superclass is invalid in the subclass



abstract classes

- It is possible to define an abstract class that contains no abstract methods.
  - **This class is used as a base class for defining new subclasses.**

abstract classes

- An object **cannot** be created from abstract class:
  - An abstract class cannot be instantiated using the **`new`** operator:

```
GeometricObject o =
    new GeometricObject();
```

  - We still define its constructors, which are invoked in the constructors of its subclasses through **constructor chaining**.
    - For instance, the constructors of **`GeometricObject`** are invoked by the constructors in the **`Circle`** and the **`Rectangle`** classes.

abstract classes as types

- An abstract class can be used as a data type:

```
GeometricObject c = new Circle(2);
```

- We can create an array whose elements are of **`GeometricObject`** type:

```
GeometricObject[] geo =
   new GeometricObject[10];
```

  - There are only **`null`** elements in the array until they are initialized with concrete objects:

```
geo[0] = new Circle();
geo[1] = new Rectangle();
…
```

The *abstract* Calendar class and its GregorianCalendar subclass

- An instance of **`java.util.Date`** represents a specific instant in time with millisecond precision
- **`java.util.Calendar`** is an abstract base class for extracting detailed information such as **`year`**, **`month`**, **`date`**, **`hour`**, **`minute`** and **`second`** from a **`Date`** object for a specific calendar
  - Subclasses of **`Calendar`** can implement specific calendar systems such as Gregorian calendar, Lunar Calendar and Jewish calendar.
    - **`java.util.GregorianCalendar`** is for the modern Gregorian calendar

The GregorianCalendar Class

- Java API for the **`GregorianCalendar`** class:
  http://docs.oracle.com/javase/8/docs/api/java/util/GregorianCalendar.html
  - **`new GregorianCalendar()`** constructs a default **`GregorianCalendar`** with the current time
  - **`new GregorianCalendar(year, month, date)`** constructs a **`GregorianCalendar`** with the specified **`year`**, **`month`**, and **`date`**
    - The **`month`** parameter is 0-based, i.e., 0 is for January, 1 is for February, …, 11 is for December.

The *abstract* Calendar class and its GregorianCalendar subclass

java.util.Calendar	
#Calendar()	Constructs a default calendar.
+get(field: int): int	Returns the value of the given calendar field.
+set(field: int, value: int): void	Sets the given calendar to the specified value.
+set(year: int, month: int, dayOfMonth: int): void	Sets the calendar with the specified year, month, and date. The month parameter is 0-based, that is, 0 is for January.
+getActualMaximum(field: int): int	Returns the maximum value that the specified calendar field could have.
+add(field: int, amount: int): void	Adds or subtracts the specified amount of time to the given calendar field.
+getTime(): java.util.Date	Returns a Date object representing this calendar's time value (million second offset from the Unix epoch).
+setTime(date: java.util.Date): void	Sets this calendar's time with the given Date object.

java.util.GregorianCalendar	
+GregorianCalendar()	Constructs a GregorianCalendar for the current time.
+GregorianCalendar(year: int, month: int, dayOfMonth: int)	Constructs a GregorianCalendar for the specified year, month, and day of month.
+GregorianCalendar(year: int, month: int, dayOfMonth: int, hour:int, minute: int, second: int)	Constructs a GregorianCalendar for the specified year, month, day of month, hour, minute, and second. The month parameter is 0-based, that is, 0 is for January.

The get Method in the Calendar Class

- The get(int field) method defined in the Calendar class is useful to extract the date and time information from a Calendar object.
- The fields are defined as constants in Calendar, as shown in the following:

Constant	Description
YEAR	The year of the calendar.
MONTH	The month of the calendar with 0 for January.
DATE	The day of the calendar.
HOUR	The hour of the calendar (12-hour notation).
HOUR_OF_DAY	The hour of the calendar (24-hour notation).
MINUTE	The minute of the calendar.
SECOND	The second of the calendar.
DAY_OF_WEEK	The day number within the week with 1 for Sunday.
DAY_OF_MONTH	Same as DATE.
DAY_OF_YEAR	The day number in the year with 1 for the first day of the year.
WEEK_OF_MONTH	The week number within the month.
WEEK_OF_YEAR	The week number within the year.
AM_PM	Indicator for AM or PM (0 for AM and 1 for PM).

```java
import java.util.*;
public class TestCalendar {
  public static void main(String[] args) {
    // Construct a Gregorian calendar for the current date and time
    Calendar calendar = new GregorianCalendar();
    System.out.println("Current time is " + new Date());
    System.out.println("YEAR:\t" + calendar.get(Calendar.YEAR));
    System.out.println("MONTH:\t" + calendar.get(Calendar.MONTH));
    System.out.println("DATE:\t" + calendar.get(Calendar.DATE));
    System.out.println("HOUR:\t" + calendar.get(Calendar.HOUR));
    System.out.println("HOUR_OF_DAY:\t" + calendar.get(Calendar.HOUR_OF_DAY));
    System.out.println("MINUTE:\t" + calendar.get(Calendar.MINUTE));
    System.out.println("SECOND:\t" + calendar.get(Calendar.SECOND));
    System.out.println("DAY_OF_WEEK:\t" + calendar.get(Calendar.DAY_OF_WEEK));
    System.out.println("DAY_OF_MONTH:\t" + calendar.get(Calendar.DAY_OF_MONTH));
    System.out.println("DAY_OF_YEAR: " + calendar.get(Calendar.DAY_OF_YEAR));
    System.out.println("WEEK_OF_MONTH: " + calendar.get(Calendar.WEEK_OF_MONTH));
    System.out.println("WEEK_OF_YEAR: " + calendar.get(Calendar.WEEK_OF_YEAR));
    System.out.println("AM_PM: " + calendar.get(Calendar.AM_PM));
    // Construct a calendar for January 1, 2020
    Calendar calendar1 = new GregorianCalendar(2020, 0, 1);
    System.out.println("January 1, 2020 is a " +
      dayNameOfWeek(calendar1.get(Calendar.DAY_OF_WEEK)) );
  }
  public static String dayNameOfWeek(int dayOfWeek) {
    switch (dayOfWeek) {
      case 1: return "Sunday"; case 2: return "Monday"; case 3: return "Tuesday";
      ... case 7: return "Saturday";
      default: return null;
    }}}
```

Interfaces

- An ***interface*** is a class-like construct that contains only abstract methods and constants.
- Why is an interface useful?
  - An interface is similar to an abstract class, but the intent of an interface is to specify behavior for objects.
    - For example: specify that the objects are comparable, edible, cloneable, …
  - Allows multiple inheritance: a class can implement multiple interfaces.

Define an Interface

- Declaration:

```
public interface InterfaceName {
   // constant declarations;
   // method signatures;
}
```

Interface Example

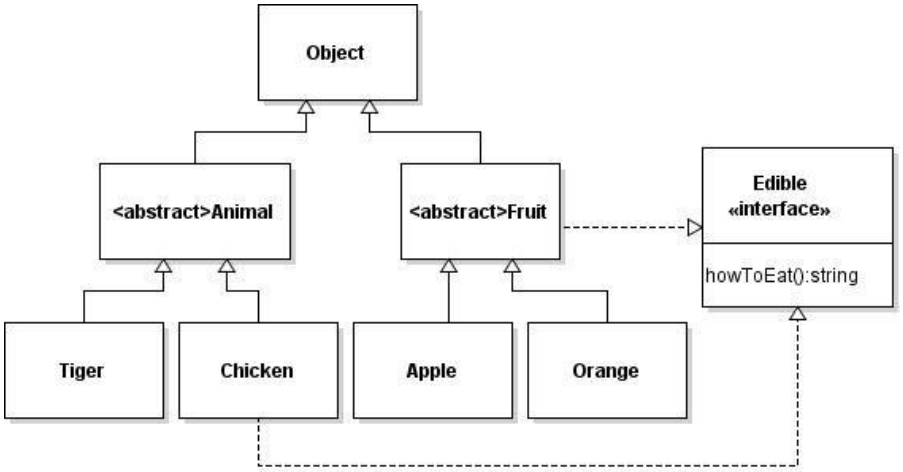
- The **Edible** interface specifies whether an object is edible

```
public interface Edible {
  public abstract String howToEat();
}
```

- The class **Chicken** implements the **Edible** interface:

```
class Chicken extends Animal implements Edible {
  public String howToEat() {
    return "Chicken: Fry it";
  }
}
```

```
interface Edible {
  public abstract String howToEat(); /** Describe how to eat */
}
abstract class Animal { }
class Chicken extends Animal implements Edible {
  public String howToEat() {
    return "Chicken: Fry it";
  }
}
class Tiger extends Animal {
}/** Does not extend Edible */
abstract class Fruit implements Edible { }
class Apple extends Fruit {
  public String howToEat() {
    return "Apple: Make apple cider";
  }
}
class Orange extends Fruit {
  public String howToEat() {
    return "Orange: Make orange juice";
  }
}
public class TestEdible {
  public static void main(String[] args) {
    Object[] objects = {new Tiger(), new Chicken(), new Apple()};
    for (Object o:objects)
      if (o instanceof Edible)
        System.out.println(((Edible)o).howToEat());
  }
}
```



Omitting Modifiers in Interfaces

- In an interface:
  - All data fields are **`public static final`**
  - All methods are **`public abstract`**
  - These modifiers can be omitted:

```
public interface T1 {
  public static final int K = 1;

  public abstract void p();
}
```

Equivalent

```
public interface T1 {
  int K = 1;

  void p();
}
```

- A constant defined in an interface can be accessed using <u>InterfaceName.CONSTANT_NAME</u>, for example: T1.K

Interfaces

- An interface is treated like a special class in Java:
  - Each interface is compiled into a separate bytecode (.class) file just like a regular class.
  - Like an abstract class, you cannot create an instance from an interface using the **`new`** operator
  - Uses of interfaces are like for abstract classes:
    - as a data type for a variable
    - as the result of casting

The Comparable Interface

- The **Comparable** interface is defined in the **java.lang** package and it is used by **Arrays.sort**

```
package java.lang;
public interface Comparable {
      int compareTo(Object o);
}
```

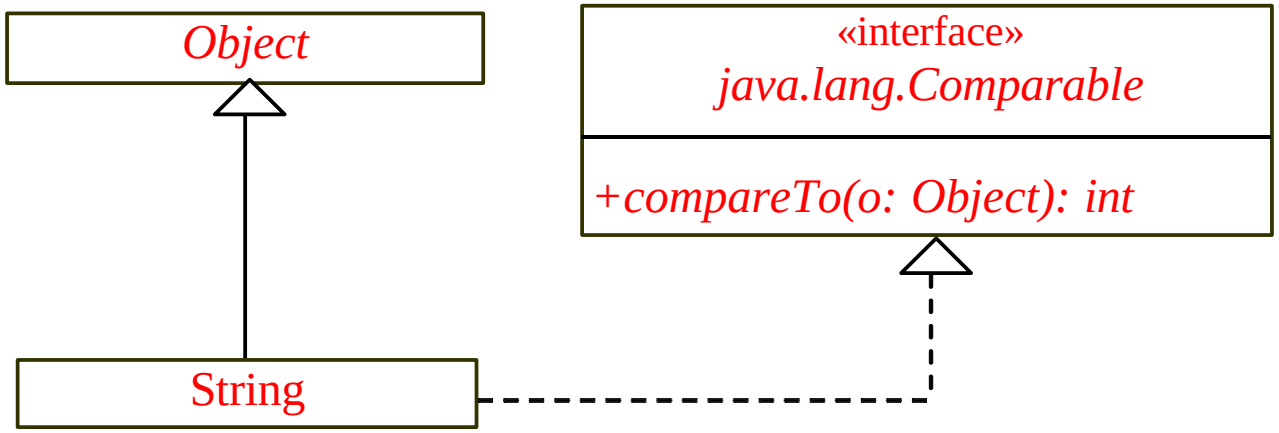
The Comparable Interface

- Many classes in the Java library implement **Comparable** (e.g., **String** and **Date**) to define a natural order for the objects:

```
public class String extends Object
    implements Comparable {
  // class body omitted
}
```

```
public class Date extends Object
    implements Comparable {
  // class body omitted
}
```

```
new String() instanceof String                    true
new String() instanceof Comparable                true
new java.util.Date() instanceof java.util.Date    true
new java.util.Date() instanceof Comparable        true
```



In UML, the interface and the methods are italicized

dashed lines and triangles are used to point to the interface

Writing a generic `max` Method

```
// Max.java: Find a maximum object
public class Max {
  /** Return the maximum of two objects */
  public static Comparable max
      (Comparable o1, Comparable o2) {
    if (o1.compareTo(o2) > 0)
      return o1;
    else
      return o2;
  }
}
```

(a)

```
// Max.java: Find a maximum object
public class Max {
  /** Return the maximum of two objects */
  public static Object max
      (Object o1, Object o2) {
    if (((Comparable)o1).compareTo(o2) > 0)
      return o1;
    else
      return o2;
  }
}
```

(b)

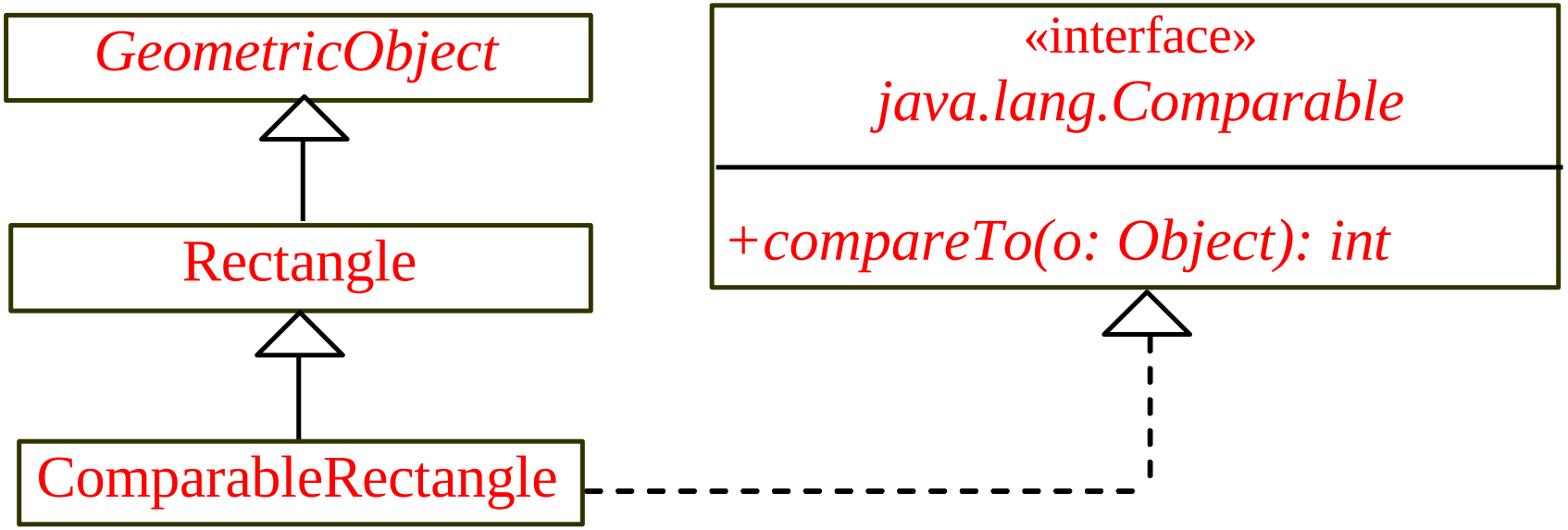
```
String s1 = "abcdef";
String s2 = "abcdee";
String s3 = (String)Max.max(s1, s2);
```

```
Date d1 = new Date();
Date d2 = new Date();
Date d3 = (Date)Max.max(d1, d2);
```

The **`return`** value from the **`max`** method is of the **`Comparable`** type. So, we need to cast it to **`String`** or **`Date`** explicitly.

Defining Classes to Implement Comparable

- We cannot use the max method to find the larger of two instances of Rectangle, because Rectangle does not implement Comparable
- We can define a new rectangle class ComparableRectangle that implements Comparable: the instances of this new class are comparable



```java
public class ComparableRectangle extends Rectangle
    implements Comparable {
  /** Construct a ComparableRectangle with specified properties */
  public ComparableRectangle(double width, double height) {
    super(width, height);
  }

  /** Implement the compareTo method defined in Comparable */
  public int compareTo(Object o) {
    if (getArea() > ((ComparableRectangle)o).getArea())
      return 1;
    else if (getArea() < ((ComparableRectangle)o).getArea())
      return -1;
    else
      return 0;
  }

  public static void main(String[] args) {
    ComparableRectangle rectangle1 = new ComparableRectangle(4, 5);
    ComparableRectangle rectangle2 = new ComparableRectangle(3, 6);
    System.out.println(Max.max(rectangle1, rectangle2));
  }
}
```

Sorting an Array of Objects

- Java provides a **`static`** **<u>`sort`</u>** method for sorting an array of **`Object`** in the **`java.util.Arrays`** class that uses the **`Comparable`** interface:

```
java.util.Arrays.sort(intArray);
```

Sorting an Array of Objects

```
public class GenericSort {
  public static void main(String[] args) {
    Integer[] intArray={new Integer(2),new Integer(4),new Integer(3)};
    sort(intArray); // or Arrays.sort(intArray);
    printList(intArray);
  }
  public static void sort(Object[] list) {
    Object currentMax;
    int currentMaxIndex;
    for (int i = list.length - 1; i >= 1; i--) {
      currentMax = list[i];
      currentMaxIndex = i; // Find the maximum in the list[0..i]
      for (int j = i - 1; j >= 0; j--) {
        if (((Comparable)currentMax).compareTo(list[j]) < 0) {
          currentMax = list[j];
          currentMaxIndex = j;
        }
      }
      list[currentMaxIndex] = list[i];
      list[i] = currentMax;
    }
  }
  public static void printList(Object[] list) {
    for (int i=0;i<list.length;i++) System.out.print(list[i]+" ");}}
```

The objects are instances of the Comparable interface and they are compared using the compareTo method.

The `Cloneable` Interface

- ***Marker Interface***: is an empty interface (does not contain constants or methods), but it is used to denote that a class possesses certain desirable properties to the compiler and the JVM.

```
package java.lang;
public interface Cloneable {
}
```

- A class that **`implements`** the **`Cloneable`** interface is marked cloneable:
  - its objects can be cloned using the **`clone()`** method defined in the **`Object`** class, and we can override this method in our classes

The `Cloneable` Interface

- **`Calendar`** (in the Java library) implements **`Cloneable`**:

```
  Calendar calendar = new GregorianCalendar(2022, 1, 1);
  Calendar calendarCopy = (Calendar)(calendar.clone());
System.out.println("calendar == calendarCopy is "
    +(calendar == calendarCopy));
```

Displays:

`calendar == calendarCopy is false`

because the references are different

```
System.out.println("calendar.equals(calendarCopy) is"
    + calendar.equals(calendarCopy));
```

`calendar.equals(calendarCopy) is true`

because the **`calendarCopy`** is a copy of **`calendar`**

Implementing the `Cloneable` Interface

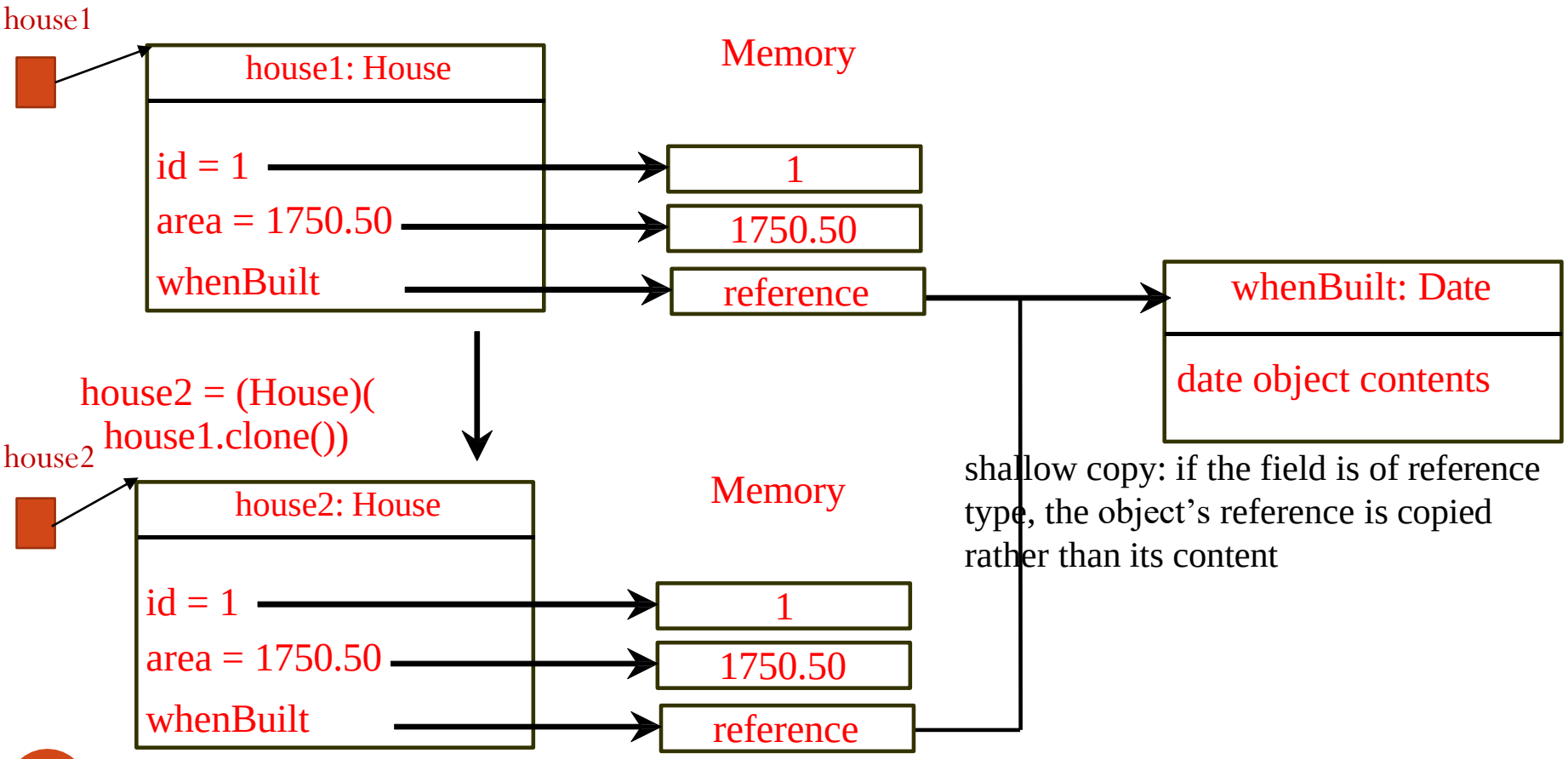
- If we try to create a clone of an object instance of a class that does not implement the **`Cloneable`** interface, it throws **`CloneNotSupportedException`**
- The **`clone()`** method in the **`Object`** class creates a new instance of the class of this object and initializes all its fields with exactly the contents of the corresponding fields of this object, as if by assignment (using a technique named ***reflection***); the contents of the reference data fields are not cloned.
  - The **`clone()`** method returns an **`Object`** that needs to be casted
- We can override the **`clone()`** method from the **`Object`** class to create custom clones

```java
public class SomethingCloneable implements Cloneable {
  public boolean equals(Object o){
    return true;
  }
  public static void main(String[] args)
      throws CloneNotSupportedException {
    SomethingCloneable s1 = new SomethingCloneable();
    SomethingCloneable s2 = (SomethingCloneable) s1.clone();
    System.out.println("s1 == s2 is " + (s1 == s2));
       // false
    System.out.println("s1.equals(s2) is " + s1.equals(s2));
       // true
  }
}
```

```java
public class House implements Cloneable, Comparable {
  private int id;
  private double area;
  private java.util.Date whenBuilt;
  public House(int id, double area) {this.id = id; this.area = area;
    whenBuilt = new java.util.Date();}
  public double getId() { return id;}
  public double getArea() { return area;}
  public java.util.Date getWhenBuilt() { return whenBuilt;}
  /** Override the protected clone method defined in the Object
    class, and strengthen its accessibility */
  public Object clone() {
    try {
      return super.clone();
    }catch (CloneNotSupportedException ex) {
      return null;
    }
  }
  /** Implement the compareTo method defined in Comparable */
  public int compareTo(Object o) {
    if (area > ((House)o).area)
      return 1;
    else if (area < ((House)o).area)
      return -1;
    else
      return 0;
  }
}
```

Shallow vs. Deep Copy

```
House house1 = new House(1, 1750.50);
House house2 = (House)(house1.clone());
```



shallow copy: if the field is of reference type, the object's reference is copied rather than its content

For ***deep copying***, we can override the clone method with custom object creation:

```
public class House implements Cloneable {
   ...
   public Object clone() { // deep copy
      try {
         House h =  (House)(super.clone());
         h.whenBuilt = (Date)(whenBuilt.clone());
         return h;
      }catch (CloneNotSupportedException ex) {
         return null;
      }
   }
   ...
}
```

For ***deep copying***, we can override the clone method with custom object creation:

```
public class House implements Cloneable {
   ...
   public Object clone() { // deep copy
        Date whenBuilt2 = (Date)(whenBuilt.clone());
        House h = new House(id, area, whenBuilt2);
        return h;
   }
   ...
}
```

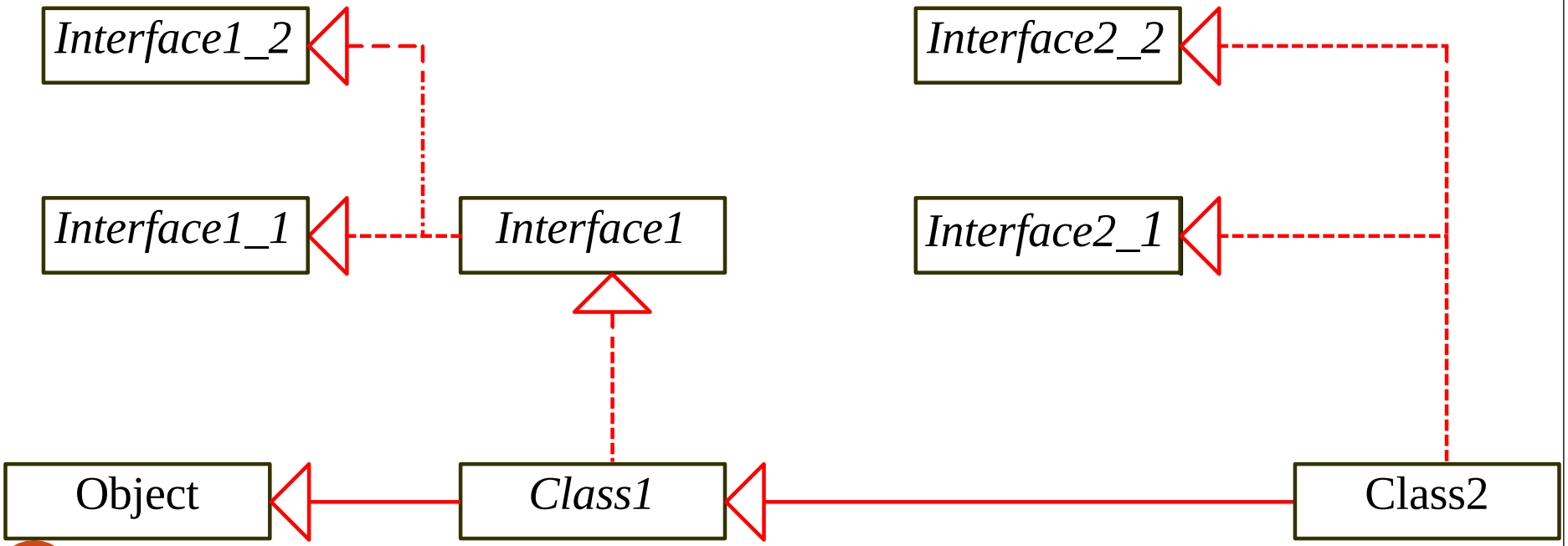
Interfaces vs. Abstract Classes

- In an interface, the data fields must be constants; an abstract class can have variable data fields
- Interfaces don't have constructors; all abstract classes have constructors
- Each method in an interface has only a signature without implementation (i.e., only abstract methods); an abstract class can have concrete methods

	Variables	**Constructors**	**Methods**
Interfaces	**All variables must be public static final**	**No constructors.** **An interface cannot be instantiated using the new operator.**	**All methods must be public abstract methods**
Abstract classes	**No restrictions**	**Constructors are invoked by subclasses through constructor chaining.** **An abstract class cannot be instantiated using the new operator.**	**No restrictions.**

Inheritance: Interfaces & Classes

- An interface can extend any number of other interfaces
- There is no root for interfaces
- A class can implement any number of interfaces



Conflicting interfaces

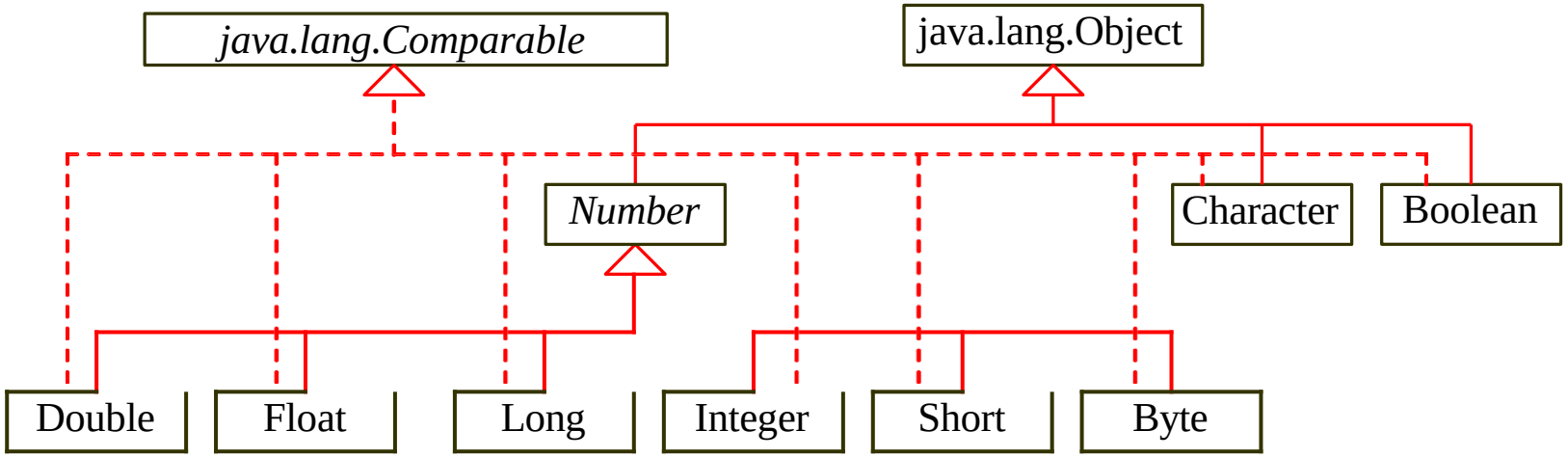
- Errors detected by the compiler:
  - If a class implements two interfaces with conflicting information, like:
    - two same constants with different values, or
    - two methods with same signature but different return type

Whether to use a class or an interface?

- *Strong is-a*: a relationship that clearly describes a parent-child relationship
  - For example: a student is a person
  - Should be modeled using class inheritance
- *Weak is-a* (or *is-kind-of*): indicates that an object possesses a certain property
  - For example: all strings are comparable, all dates are comparable
  - Should be modeled using interfaces
- You can also use interfaces to circumvent single inheritance restriction if multiple inheritance is desired

Wrapper Classes

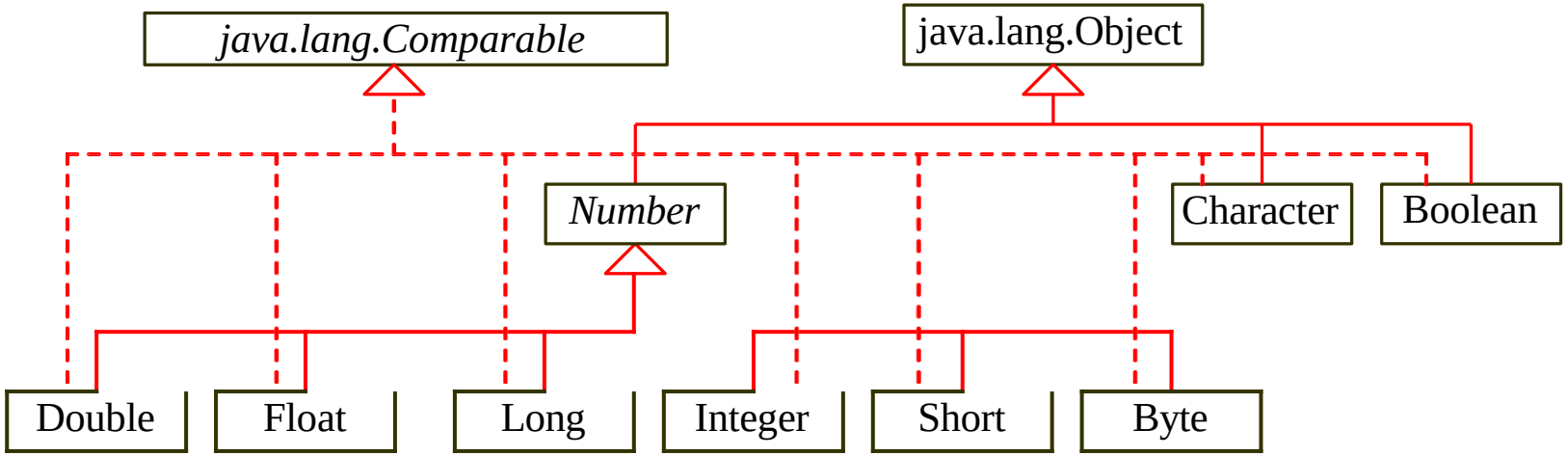
- Primitive data types in Java ➔ **Better performance**
  - However, data structures (ArrayList) expect objects as elements
- Each primitive type has a wrapper class: Boolean, Character, Short, Byte, Integer, Long, Float, Double



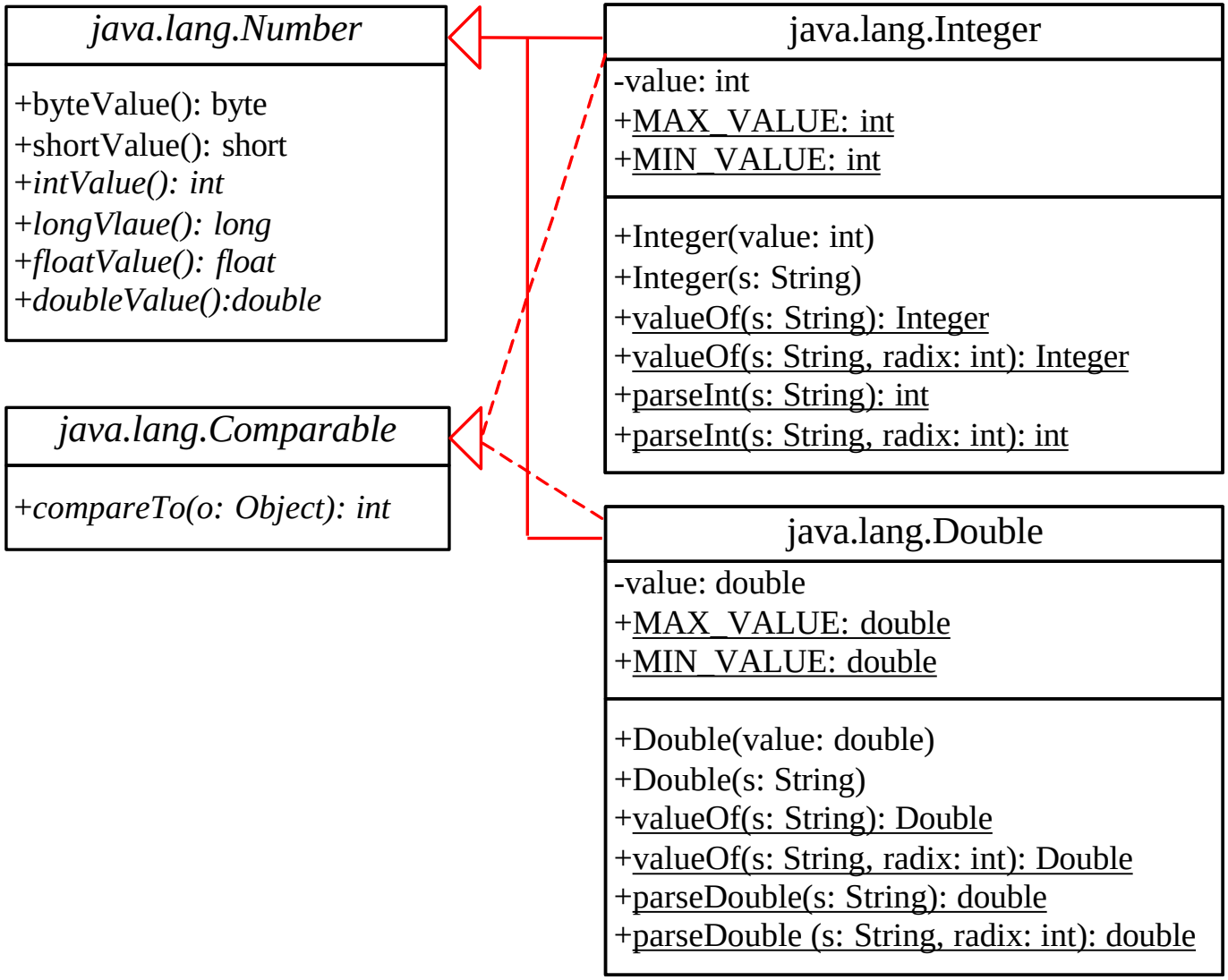
  - The wrapper classes do not have no-arg constructors
  - The instances of all wrapper classes are immutable: their internal values cannot be changed once the objects are created

Wrapper Classes

- Each wrapper class overrides the **`toString`** and **`equals`** methods defined in the **`Object`** class
- Since these classes implement the **`Comparable`** interface, the **`compareTo`** method is also implemented in these classes



The `Integer` and `Double` Classes



The `static` `valueOf` methods

- The numeric wrapper classes have a **`static`** method **`valueOf(String s)`** to create a new object initialized to the value represented by the specified string:

```
Double doubleObject = Double.valueOf("12.4");

Integer integerObject = Integer.valueOf("12");
```

- Each numeric wrapper class has overloaded parsing methods to parse a numeric string into an appropriate numeric value:

```
double d = Double.parseDouble("12.4");

int i = Integer.parseInt("12");
```

Wrapper Classes

- Automatic Conversion Between Primitive Types and Wrapper Class Types:
  - Since JDK 1.5, Java allows primitive type and wrapper classes to be converted automatically:
    - ***boxing*** of primitive types into wrapper types when objects are needed

```
Integer[] intArray = {2, 4, 3};
```

Equivalent

```
Integer[] intArray = {new Integer(2),
  new Integer(4), new Integer(3)};
```

    - ***unboxing*** of wrapper types into primitive types when primitive types are needed

```
int n = intArray[0] + intArray[1] + intArray[2];
```

Unboxing

`BigInteger` and `BigDecimal`

- **`BigInteger`** and **`BigDecimal`** classes in the **`java.math`** package:
  - For computing with very large integers or high precision floating-point values
    - **`BigInteger`** can represent an integer of any size
    - **`BigDecimal`** has no limit for the precision (as long as it's finite=terminates)
  - Both are *immutable*
  - Both extend the ***`Number`*** class and implement the **`Comparable`** interface.

BigInteger and BigDecimal

```
BigInteger a = new BigInteger("9223372036854775807");
BigInteger b = new BigInteger("2");
BigInteger c = a.multiply(b); // 9223372036854775807 * 2
System.out.println(c);
```

```
18446744073709551614
```

```
BigDecimal a = new BigDecimal(1.0);
BigDecimal b = new BigDecimal(3);
BigDecimal c = a.divide(b, 20, BigDecimal.ROUND_UP);
System.out.println(c);
```

```
0.33333333333333333334
```

BigInteger and BigDecimal

```
import java.math.*;
public class LargeFactorial {
  public static void main(String[] args) {
    System.out.println("50! is \n" + factorial(50));
  }
  public static BigInteger factorial(long n) {
    BigInteger result = BigInteger.ONE;
    for (int i = 1; i <= n; i++)
      result = result.multiply(new BigInteger(i+""));
    return result;
  }
}
```

```
30414093201713378043612608166064768844377641
568960512000000000000
```